

AVINASH MISHRA

AI-First Senior Technical Program Manager | Agentic AI & Loop Engineering Leadership | GenAI/LLM Platform Delivery | AI Governance & Cloud FinOps

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Available for U.S.-based remote roles — Full-Time or Contract • No visa sponsorship required • Full workday overlap with U.S. business hours

25,000+ Active Users | >90% RAG Faithfulness | 99.95% Uptime | 2–3x AI-Assisted Engineering Output | 20–30% Cloud Cost Savings

PROFESSIONAL SUMMARY

AI-first Senior Technical Program Manager with 10+ years directing enterprise SaaS and, most recently, GenAI/LLM platform delivery for a multi-tenant system serving 22 institutional clients and 25,000+ end users. Leads by putting AI-assisted engineering, AI workflow automation, and AI governance at the center of program execution — using modern AI capabilities to compound engineering velocity rather than relying on hands-on coding depth. Owns program-level Loop Engineering: governing the iterative, closed-loop AI evaluation and continuous-optimization workflow (Ragas, G-Eval, LLM-as-a-Judge; golden datasets; quality gates) that sustains >90% output faithfulness, and driving enterprise-wide AI adoption and governance standards across the engineering organization. Also directed a semantic-caching architecture that cut inference latency and cost, and a bi-weekly FinOps governance model — partnering closely with engineering leadership on architecture strategy and technical trade-offs while owning roadmap, risk, delivery telemetry (DORA), and executive/customer communication across a 40–45 person distributed engineering matrix. Seeking to bring this AI-enabled delivery discipline to a larger global engineering organization.

CORE COMPETENCIES

AI Leadership: Agentic AI • Multi-Agent Systems • Loop Engineering • Closed-Loop AI Evaluation • AI Observability • Model Context Protocol (MCP) • AI-Assisted SDLC • AI Workflow Automation • AI Engineering Productivity • AI Governance & Responsible AI • LLMOps • Prompt Engineering • Multi-Model Routing • AI Cost Optimization • AI Center of Excellence (CoE) • Enterprise AI Adoption • AI ROI & Business Value Realization • Azure AI Foundry

Program Leadership: Technical Program Management • Program & Portfolio Strategy • Executive Stakeholder Management • Cross-Functional Leadership • Global/Distributed Team Leadership • Roadmap Planning • Risk & Dependency Management • Change Management • Release Governance • Agile Delivery & Scrum • DORA Metrics

Cloud & Platform: Enterprise SaaS • Multi-Tenant Architecture • Cloud FinOps • AWS • Azure • GCP • Kubernetes • Microservices • REST APIs

Engineering Productivity: Cursor • GitHub Copilot • Sourcegraph Cody • Amplitude • AI-Assisted Engineering • Delivery Analytics

PROFESSIONAL EXPERIENCE

Senior Technical Program Manager & Founder — Noetic Education Pvt. Ltd.

Jun 2016 – Present • Mumbai, India
Directs a bootstrapped, self-funded B2B enterprise EdTech platform (founded 2016), governing 5–8 concurrent programs and a 40–45 person distributed engineering matrix delivering 60+ software systems for 22 enterprise clients, 500+ business users and 25,000+ active end users. Platforms are deployed white-label under each client's own brand by design — protecting institutional goodwill rather than building a public consumer brand — including a premium white-label add-on, StudyAtom, licensed within the existing client base.

Enterprise GenAI & RAG Transformation | 2024 – 2026

- AI platform strategy:** directed the program establishing GenAI/LLM capability as a reusable, model-agnostic orchestration layer — rather than isolated features — across the full multi-tenant client base, with capability-based multi-model routing (Claude Haiku/Sonnet, GPT-4o) and automatic failover on provider rate-limits.
- Loop Engineering — AI governance & evaluation:** owned program governance of a closed-loop, iterative AI evaluation and continuous-optimization workflow — defining AI quality gates, governing the golden dataset and LLM-as-a-Judge evaluation program (Ragas, G-Eval) gated in CI/CD, and prioritizing prompt-, retrieval-, and model-routing optimizations from evaluation results — measuring faithfulness, context precision and answer relevance to enforce production-readiness before release; sustained >90% output faithfulness, with monthly Cohen's Kappa calibration (>0.8) to control judge bias.
- Latency & cost engineering:** set technical strategy for a semantic-caching architecture (vector similarity) achieving a 30–40% cache-hit rate, cached TTFT p95 <500ms and ITL p95 <50ms, including directing root-cause analysis with engineering on an SSE-buffering load-balancer defect.
- Responsible AI:** drove adoption of PII-tokenization and mandatory source-citation guardrails into the SDLC, turning production AI output into auditable, source-grounded answers rather than unverified text.

- **AI cost optimization / FinOps:** governed model routing, semantic caching and token-usage optimization as commercial-sustainability decisions via bi-weekly FinOps reviews — identifying 20–30% cloud cost savings and >60% spot/savings-plan utilization, with per-feature/tenant unit-economics reporting down to \$0.15 per processed transaction for a high-volume feature.
- **Delivery telemetry:** instituted DORA-based metrics (commit lead time <24–48 hrs, change failure rate <15%, 2–4 week feature lead time) and a 70/20/10 features/tech-debt/maintenance capacity model.
- **AI-assisted engineering productivity:** drove team-wide rollout of AI-assisted coding tools (Cursor, GitHub Copilot, Sourcegraph Cody) across the engineering matrix by 2025 — an AI-first workflow-automation initiative that lifted overall engineering output 2–3x and compounded the DORA velocity gains above.

Premium Add-On Productization — StudyAtom (White-Label) | 2022 – 2023

- **Premium upsell line:** directed development and rollout of StudyAtom, a white-label digital-learning add-on (course catalogue, content delivery, assessments, progress tracking) licensed as a premium tier to the existing 22-client base — expanding wallet share without a public consumer brand or channel conflict with clients.
- **Architecture governance:** partnered with engineering leadership to shape the decision to decouple the add-on module from the core ERP/LMS into a shared headless microservices backend with independent deployment pipelines — improving add-on engineering velocity (up to 5 deploys/day) while preserving enterprise stability and QA-gated bi-weekly release governance for core systems.
- **Governance maturity:** matured program governance (quarterly roadmap planning, cross-functional dependency tracking, executive reviews) spanning Engineering, Product, Academic Content, Client Success and Support.

High-Availability Platform Scaling | 2019 – 2021

- **Crisis-driven scale-up:** directed platform-wide scaling through COVID-driven demand surge, growing the platform from 5 to 60+ software systems and from ~2,000 to 25,000+ active end users across 22 institutional clients, while sustaining a 99.95% availability target.
- **Incident response leadership:** directed cross-functional incident response and root-cause analysis on database connection-limit exhaustion under concurrent-user spikes, prioritizing engineering effort and governing rollout planning for a zero-downtime remediation using centralized connection pooling (PgBouncer) and asynchronous read-replica offload.
- **Unified operations:** directed consolidation of disconnected academic workflows (admissions, attendance, exams, communication) into one digital platform, governing release planning across multiple concurrent releases delivered with no service degradation.

Enterprise Platform Modernization & Multi-Tenant SaaS Migration | 2016 – 2018

- **Program leadership:** directed a multi-year platform modernization program transitioning fragmented, spreadsheet-driven institutional workflows (admissions, attendance, fees, exams) into a unified, multi-tenant SaaS platform — replacing per-client custom builds with one reusable foundation; grew the initial engineering function to deliver the first 5 core systems, supporting up to 2,000 active end users.
- **Architecture direction:** set architecture direction for institution-level data isolation, configurable workflows and API-first integrations, establishing the reusable platform foundation that scaled 12x in systems and users during the COVID growth phase.
- **Agile governance:** introduced structured Agile delivery (sprint planning, backlog refinement, risk/dependency reviews) across the engineering function, sustaining an 80–85% sprint say-do ratio.

TECHNICAL FLUENCY

AI / GenAI: Agentic AI, Loop Engineering, Model Context Protocol (MCP), Azure AI Foundry, LLM Orchestration & Multi-Model Routing, RAG Architecture, Prompt Engineering, Ragas, G-Eval, LLM-as-a-Judge, Semantic Caching, Vector Search, Anthropic API, OpenAI API (model routing & evaluation)

Cloud & Infrastructure: AWS, Azure, GCP, Firebase, PostgreSQL, MySQL, PgBouncer, Read Replicas, Spot/Savings Plans, Kubernetes, Microservices & REST APIs

Delivery & Governance: Agile, Scrum, Kanban, DORA Metrics, Multi-Tenant SaaS Architecture, Release & Risk Management, Cloud FinOps (CAPEX/OPEX)

AI-Assisted Engineering: Cursor, GitHub Copilot, Sourcegraph Cody, Amplitude, AI Workflow Automation, Delivery Analytics

Collaboration Tools: Jira, Confluence, Git, GitHub, GitLab, Slack, Microsoft Teams, Notion

EDUCATION

Bachelor of Science, Industrial Science — 2013

Aircraft Maintenance Engineering (Avionics) — 2013